ERTS Lab
Department of Computer Science and Engineering
Indian Institute of Technology Bombay, Powai
Mumbai - 400 076.

Engineering a better tomorrow

Certificate of Completion

This is to certify that **Abhinav Goel**, a student of **PDPM Indian Institute of Information Technology, Design and Manufacturing, Madhya Pradesh** has participated in the **e-Yantra Robotics Competition (eYRC 2025-26)**.

He/She is a member of the team having the following team members,

1. Abhinav Goel
2. Harshit Kumar Saxena
3. Shirshendu Ranjana Tripathi
4. Akash Dhyani

This team has successfully completed all the assigned tasks in **KrishiCobot** theme.

Prof. Shivaram Kalyanakrishnan
Principal Investigator, e-Yantra
Associate Professor
Department of Computer Science and Engineering
Indian Institute of Technology Bombay

EYRC2025-261776172329.8123SEBKEGGV

e-Yantra is a project sponsored by MHRD, Government of India, under the National Mission on Education through ICT (NMEICT).

[P.T.O]

Theme: *KrishiCobot*

Theme Description:

In this theme, teams developed an autonomous smart farm management system using a UR5 robotic arm and eBot (mobile robot). The UR5 autonomously handled fertilizer containers by placing them onto the mobile robot and disposing of empty containers, while also performing vision-based inspection of harvested produce on a conveyor to identify and remove defective fruits. The mobile robot navigated the farm environment using LiDAR-based autonomous navigation, identified plant conditions using geometric structures placed, and delivered fertilizer where required. It returned to the robotic arm for container exchange. The system required coordinated integration of perception, autonomous navigation, and robotic manipulation to achieve efficient and reliable operation.

In stage 1 of the competition, teams implemented this theme in a simulator (Gazebo), and in stage 2, they got remote access to perform the task on real industrial hardware consisting of a UR5 robotic arm and eBot deployed at IIT Bombay.

Theme Learning: *Robot Operating System 2 (ROS 2), Gazebo Simulator, Computer Vision, Autonomous Navigation, LiDAR-based feature detection, Joint and Velocity-based Servoing for Manipulation, RViz 2, Multi-robot collaboration, and Algorithm Development using Python and Git.*

Certificate Level: The team has been awarded **Level 2** certificate in the e-Yantra Robotics Competition (eYRC).

> ***Level 2 indicates:*** *The team demonstrated a clear understanding of the problem statement via a full implementation and was in top 10-30 ranked team in the competition.*

Certificate Level Matrix

Certificate Level	Certificate Description
Level 1	**Certificate of Merit**
Level 2	**Certificate of Completion**
Level 3	Certificate of Participation
Level 4	Letter of Participation

* Level 1 > Level 2 > Level 3 > Level 4 (Level 1 is highest achievement)